

Lab 09 Relationships

NOTE: Use the project from example of week 09.

1. Update **ShopController** to be the same behavior as **ProductController**.

DB - Shop: List

Ask Gemini

localhost:8000/shops

Product

Shop

Shop: List

Search

Term

Filter

Clear

Create Shop

< 1 2 3 >

List of Shops

Code	Name	Owner	No. of Products
SH001	Pachara Shop	Pachara	2
SH002	Phasit Shop	Phasit	3
SH003	Other Shop	Pachara	3
SH101	Adam Shop	Adam	3
SH102	MadA Shop	Adam	1

© Copyright Pachara's Database.

Product Shop

Shop: SH001

View Products Update Delete

Code :: SH001
Name :: Pachara Shop
Owner :: Pachara
Location :: 18.80045380, 98.94889980
Address :: College of Arts, Media and Technology
239 Huaykaew Rd., Suthep,
Mueang Chiang Mai District, Chiang Mai 50200

© Copyright Pachara's Database.

Product Shop

Shop: SH001: Products

Search

Term

Min Price

Max Price

Filter Clear

< Back

List of Products for Pachara Shop

Code	Name	Price	No. of Shops
PD001	Programming PHP	345.00	2
PD101	Modern PHP Web Development	758.55	4

© Copyright Pachara's Database.

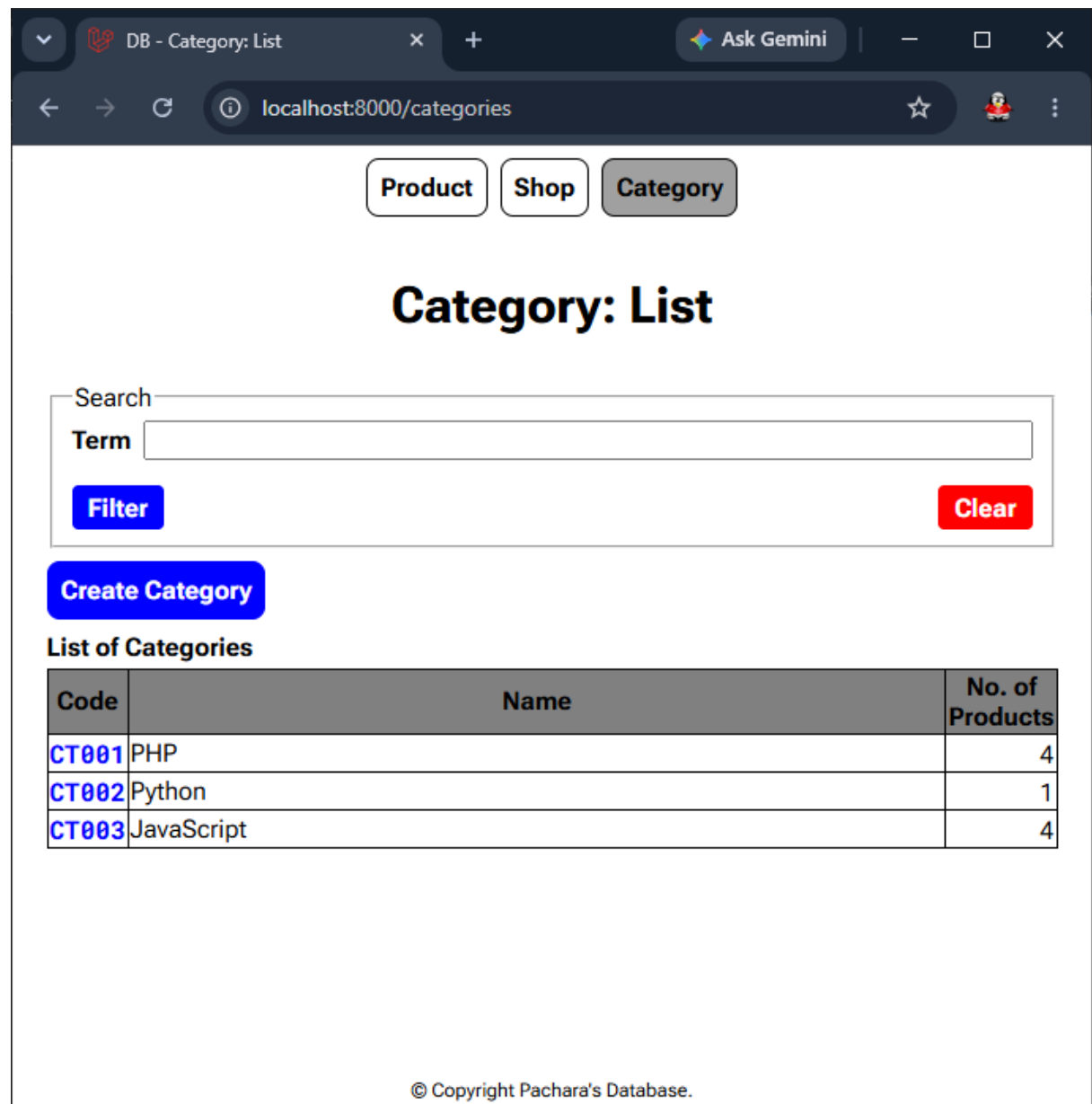
The search criteria of this route are same as route `products.list` that can search by search term, min price and max price. **Don't create a new one**, just reuse from `ProductController`.

2. Create **Category** model that has relation *one-to-many* with **Product** model.

Extract the given file **lab09-migrations.zip** to **database/migrations**. Then run the following command.

```
php artisan migrate
```

Create **Category** model and **CategoryController**. Then create **category routes** that are the same as **shop routes**.



DB - Category: List

Ask Gemini

localhost:8000/categories

Product Shop Category

Category: List

Search Term

Filter Clear

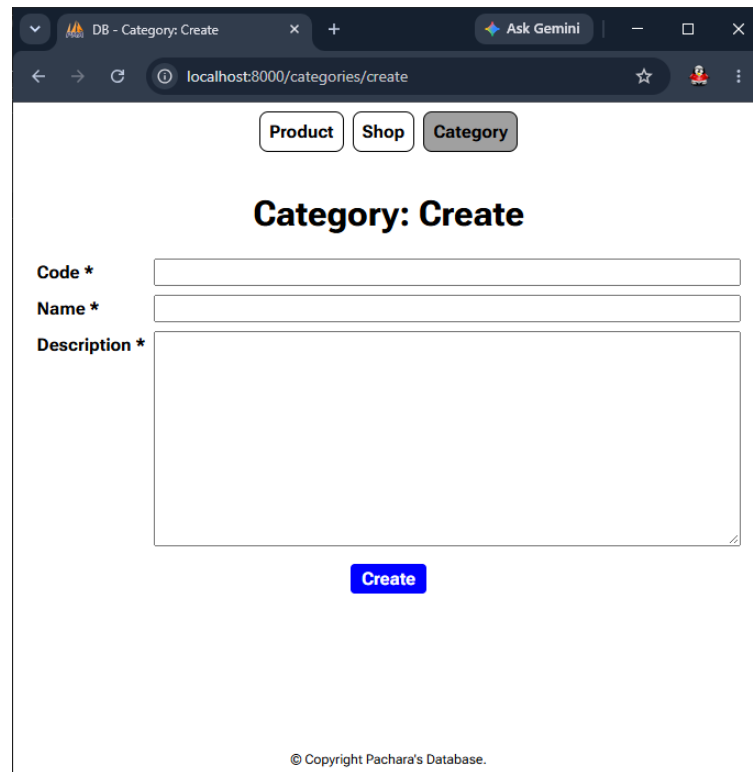
Create Category

List of Categories

Code	Name	No. of Products
CT001	PHP	4
CT002	Python	1
CT003	JavaScript	4

© Copyright Pachara's Database.

This route can search by using **search term** on **code** and **name**.



DB - Category: Create

localhost:8000/categories/create

Product Shop Category

Category: Create

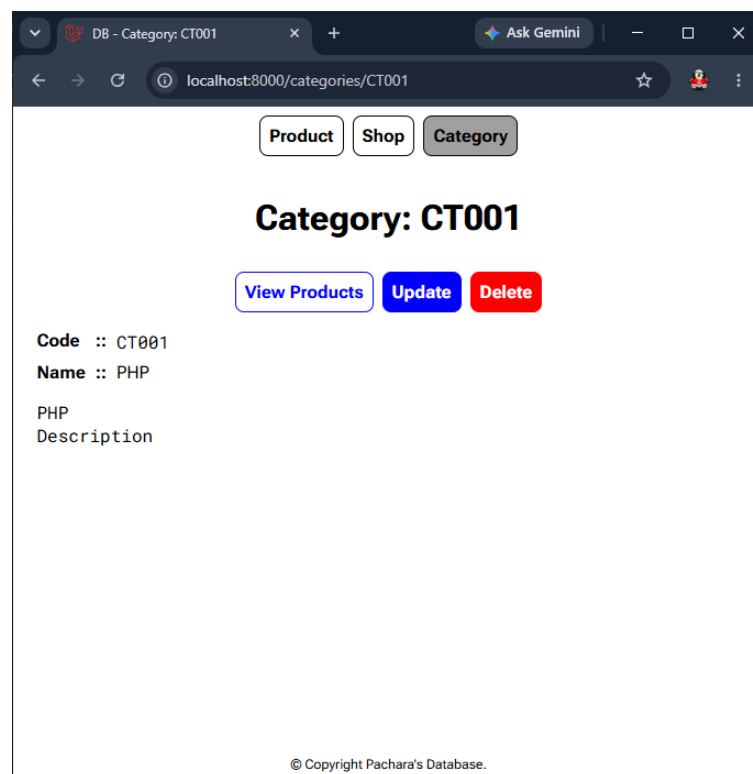
Code *

Name *

Description *

Create

© Copyright Pachara's Database.



DB - Category: CT001

localhost:8000/categories/CT001

Product Shop Category

Category: CT001

View Products Update Delete

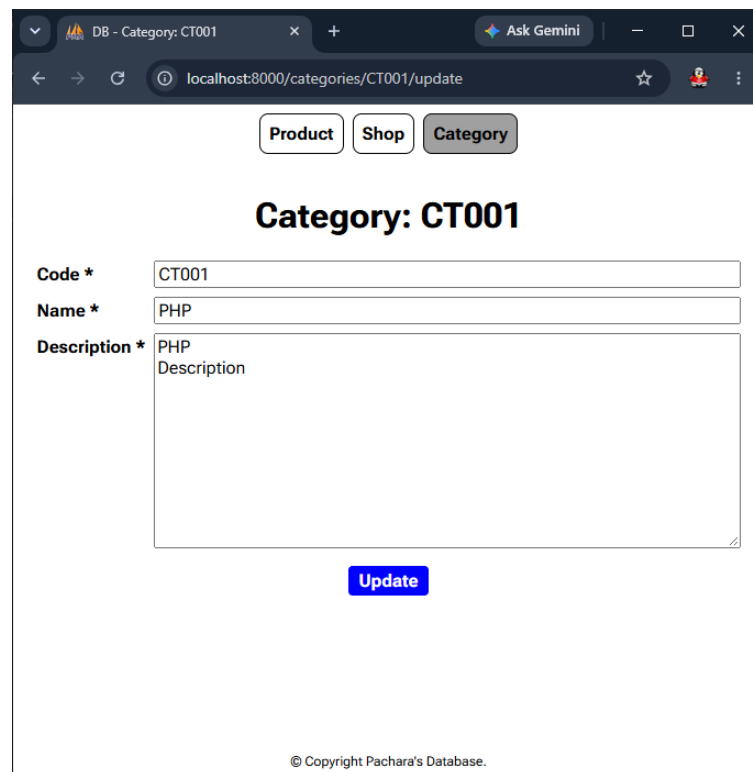
Code :: CT001

Name :: PHP

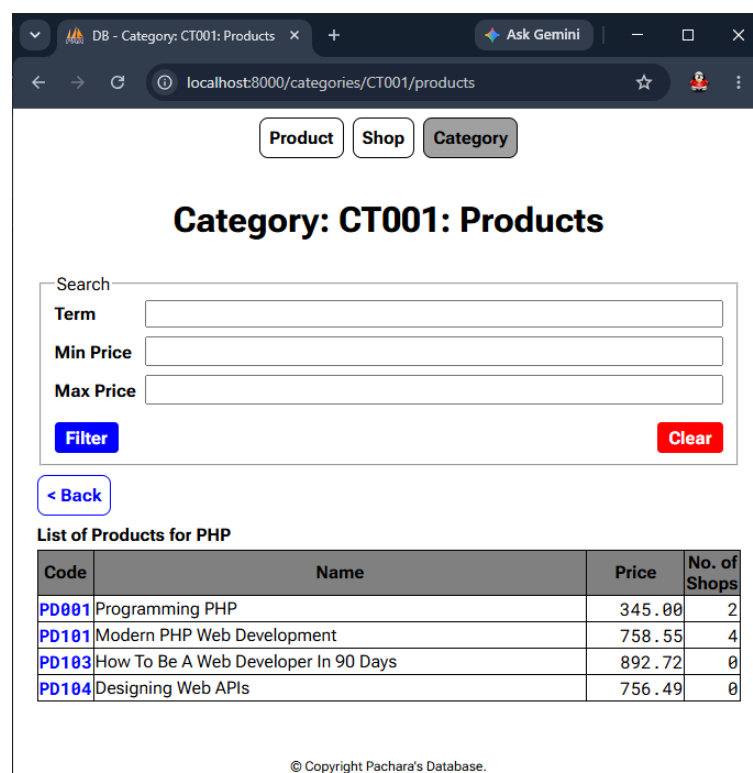
PHP

Description

© Copyright Pachara's Database.



The screenshot shows a web browser window with the URL `localhost:8000/categories/CT001/update`. The page has a navigation bar with buttons for 'Product', 'Shop', and 'Category'. The main heading is 'Category: CT001'. Below the heading, there are three form fields: 'Code *' with the value 'CT001', 'Name *' with the value 'PHP', and 'Description *' with the value 'PHP Description'. A blue 'Update' button is located below the description field. At the bottom of the page, there is a copyright notice: '© Copyright Pachara's Database.'



The screenshot shows a web browser window with the URL `localhost:8000/categories/CT001/products`. The page has a navigation bar with buttons for 'Product', 'Shop', and 'Category'. The main heading is 'Category: CT001: Products'. Below the heading, there is a search section with three input fields: 'Search Term', 'Min Price', and 'Max Price'. A blue 'Filter' button is located below the 'Max Price' field, and a red 'Clear' button is located to the right of the 'Filter' button. Below the search section, there is a '< Back' button. The main content area is titled 'List of Products for PHP' and contains a table with the following data:

Code	Name	Price	No. of Shops
PD001	Programming PHP	345.00	2
PD101	Modern PHP Web Development	758.55	4
PD103	How To Be A Web Developer In 90 Days	892.72	0
PD104	Designing Web APIs	756.49	0

At the bottom of the page, there is a copyright notice: '© Copyright Pachara's Database.'

The search criteria are the same as route `products.list` that can search by **search term**, **min price** and **max price**.

3. Update all views that display **product data** except **categories.view-products** to display **category data**.

Update route `products.list`, `products.view` and `shops.view-products` to show category and **handle N+1 query problem** in collection retrievals.

Product: List

Search

Term

Min Price

Max Price

Filter

Create Product

< 1 2 >

List of Products

Code	Name	Category	Price	No. of Shops
PD001	Programming PHP	PHP	345.00	2
PD002	JavaScript: The Definitive Guide	JavaScript	250.00	3
PD003	Learning PHP, MySQL & JavaScript	JavaScript	450.00	2
PD004	Python Crash Course, 2nd Edition	Python	560.00	1
PD101	Modern PHP Web Development	PHP	758.55	4

© Copyright Pachara's Database.

route `products.list`

Product: PD001

View Shops Update Delete

Code :: PD001

Name :: Programming PHP

Category :: [CT001] PHP

Price :: 345.00

Why is PHP the most widely used programming language on the web?
 This updated edition teaches everything you need to know to create effective web applications using the latest features in PHP 7.4. You'll start with the big picture and then dive into:
 language syntax
 programming techniques
 and other details
 using examples that illustrate both correct usage and common idioms.

© Copyright Pachara's Database.

route `products.view`

Product Shop Category

Shop: SH002: Products

Search

Term

Min Price

Max Price

Filter Clear

< Back

List of Products for Phasit Shop

Code	Name	Category	Price	No. of Shops
PD001	Programming PHP	PHP	345.00	2
PD002	JavaScript: The Definitive Guide	JavaScript	250.00	3
PD101	Modern PHP Web Development	PHP	758.55	4

© Copyright Pachara's Database.

route **shops.view-products**

Note: For `products.create-form` and `products.update-form`, we **don't touch them for a while**. Because we have to learn about **relationships manipulation** in the next chapter.